

1. Core Messaging & Real-Time (The Essentials)

These features form the "heart beat" of your application.

- **One-to-One Messaging:** Low-latency text exchange using Socket.io.
- **Typing Indicators:** Real-time "User is typing..." feedback.
- **Presence Indicators:** Visual cues for "Online," "Offline," and "Last Seen" timestamps.
- **Message Statuses:** Visual checkmarks for **Sent** (one tick), **Delivered** (two ticks), and **Seen** (blue ticks).
- **Delete/Edit Messages:** The ability to "Delete for Everyone" or edit a typo after sending.

2. Multimedia & File Handling

Since you're using MERN, these will require a cloud storage provider like **Cloudinary** or **AWS S3**.

- **Image & Video Sharing:** In-chat preview with "tap to expand" functionality.
- **Document Support:** Sending PDFs, .docx, and other files with download buttons.
- **Audio Messages:** A "Push-to-Talk" voice recorder with a waveform visualizer.
- **Link Previews:** Automatically fetch and display metadata (title, image, description) when a user pastes a URL.

3. Communication (Calls)

- **Audio Calling:** Peer-to-peer voice calls using WebRTC (or an SDK like ZegoCloud).
- **Video Calling:** High-quality video calls with a "Switch Camera" and "Mute" toggle.
- **Call History:** A dedicated tab to see missed, incoming, and outgoing calls.

4. User Profile & Discovery

- **Advanced Search:** Search for users by username or display name using MongoDB text indexes.
- **Profile Customization:** Uploading an avatar, setting a "Bio," and choosing a display name.
- **Blocking System:** A robust privacy feature to prevent specific users from messaging or calling you.
- **Contacts Integration:** (Optional) If building for mobile, the ability to sync phone contacts who also use the app.

5. Engagement & Advanced Features

These features make your app stand out and feel like a modern 2026 product.

- **Ephemeral Content (Stories):** Status updates that vanish after 24 hours (using MongoDB TTL indexes).

- **Disappearing Messages:** A chat-specific setting where messages delete themselves after a set time (e.g., 7 days).
- **Message Reactions:** Using emojis (❤️, 👍, 😊) to react to specific message bubbles.
- **Push Notifications:** Alerts via Firebase (FCM) so users get notified even when the app is closed.
- **AI Moderation:** (A perfect fit for your interests!) Automatically flagging or blurring inappropriate images and toxic text before they are displayed.